Aggregate Load Inertia from the Composite Load Model

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***Abstract*—**

***Index Terms*—composite load model, inertia constant, load inertia, frequency stability, inverter-based resources, inertia estimation**

I. Introduction

Power systems are experiencing declining synchronous inertia as IBR penetration grows. Accurate inertia quantification is critical for frequency stability assessment. Load inertia (from induction motors) contributes non-trivially but is frequently ignored or lumped into generator estimates.

The WECC Composite Load Model (CLM) is the industry-standard model for dynamic load representation and is widely deployed in stability studies. It contains explicit inertia constants (H) for three-phase induction motor sub-models (Motors A, B, C). However, no formal method exists to extract an aggregate load inertia estimate from the model — the inertia content is implicit in the parameters but never computed.

This paper makes the following contributions: (i) derives a closed-form expression for aggregate load inertia H_{load} as a function of load composition, rules of association, and motor H values; (ii) characterises sensitivity of H_{load} to uncertainty in motor H parameters and load composition; (iii) identifies Motor D as a structural gap (performance model, no H), bounds the inertia it omits, and shows the gap is small and shrinking as residential air-conditioning electronifies; (iv) demonstrates the method on [test case].

II. The Composite Load Model

A. Model Structure

B. Induction Motor Sub-models (Motors A, B, C)

Motors A, B, and C are aggregated three-phase induction-motor models that share a common fifth-order structure [?]. Four of the states represent the rotor-flux dynamics through the d- and q-axis transient and subtransient internal EMFs E'_d, E'_q, E''_d, E''_q (EPRI Eqs. A-1–A-4), which are governed by the transient and subtransient rotor time constants T_{p0} and T_{pp0} together with the synchronous, transient, and subtransient reactances L_s, L_p, and L_{pp} and the stator resistance R_a. The fifth state is the rotor slip s, whose dynamics constitute the only equation in which the motor inertia constant appears [?], [?]:

$$\frac{ds}{dt} = -\frac{\left(E''_d i_d + E''_q i_q\right) - T_m}{2H} \tag{1}$$

Equation (1) is the machine swing equation $2H\,\dot{s} = T_m - T_e$, where $T_e = E''_d i_d + E''_q i_q$ is the per-unit electromagnetic (air-gap) torque and T_m is the mechanical load torque, which follows the speed-dependent characteristic [?]

$$T_m = T_{m0}\, w^{E_{trq}}, \qquad w = 1 - s, \tag{2}$$

with w the per-unit rotor speed. The terms are:

- s — rotor slip (pu), $s = (\omega_s - \omega_r)/\omega_s$, the per-unit deviation of rotor speed ω_r from synchronous speed ω_s (the single mechanical state); $w = 1 - s$ is the per-unit rotor speed;
- E''_d, E''_q — d- and q-axis subtransient internal EMFs (pu), the rotor-flux states governed by EPRI Eqs. A-3–A-4;
- i_d, i_q — d- and q-axis components of the stator terminal current (pu), obtained algebraically from the terminal voltage and the subtransient EMFs through the stator network (EPRI Eqs. A-6–A-7);
- T_{m0} — rated mechanical torque (pu on the motor MVA base), set at initialisation to the steady-state electromagnetic torque $T_{m0} = E''_{d0} i_{d0} + E''_{q0} i_{q0}$;
- E_{trq} — speed exponent of the mechanical load torque [?]; it sets how the load torque varies with rotor speed (e.g. $E_{trq} \approx 0$ for constant-torque loads such as Motor A, $E_{trq} \approx 2$ for the variable-torque fan and pump loads of Motors B and C);
- ω_0 — synchronous (electrical) frequency, which scales the slip-frequency terms in the EMF state equations (EPRI Eqs. A-1–A-4);
- H — motor inertia constant (s), the ratio of kinetic energy stored at synchronous speed to the motor MVA base, as defined in (3).

The motor MVA base is fixed at initialisation from the active load assigned to the motor and its loading factor LF (Section III). Crucially, the inertia constant H enters the model *only* through (1): it is the sole parameter encoding rotational inertia in Motors A, B, and C, and the electrical sub-model carries no inertia term. Table I lists the NERC-recommended inertia constants for the three motor types.

TABLE I
NERC-recommended inertia constants H (s) for the three-phase induction-motor sub-models [?].

Motor	Representative loads	Comm.	Ind.
A	HVAC compressors, pumps	0.10	0.15
B	Large fans, compressors	0.50	0.80
C	Centrifugal pumps	0.10	0.80

C. Motor D

D. Load Composition and Rules of Association

III. Aggregate Load Inertia from the Composite Load Model

A. Kinetic Energy of a Single Motor Sub-model

For a rotating machine with moment of inertia J ($\mathrm{kg\,m^2}$) and synchronous mechanical speed ω_s ($\mathrm{rad\,s^{-1}}$), the inertia constant H is defined as the ratio of stored kinetic energy at synchronous speed to the machine's rated apparent power S_B:

$$H = \frac{E_k}{S_B} = \frac{\frac{1}{2}J\omega_s^2}{S_B} \tag{3}$$

where ω_s is related to the nominal system frequency f_0 (50 Hz in the Australian NEM) by the machine's pole count. So defined, H has units of seconds and is independent of pole number [**?**]. The motor speed and torques in (1) are expressed in per unit on this same synchronous-speed base, so the H of (3) is exactly the inertia constant that appears in the slip equation (1).

B. Aggregate Kinetic Energy at a Load Bus

For a set of rotating machines that are mechanically independent (i.e., do not share a common shaft), the total stored kinetic energy is the scalar sum of individual contributions [**?**]:

$$E_{k,\mathrm{bus}} = \sum_i E_{k,i} = \sum_i H_i S_{B,i} \tag{4}$$

At a CLM load bus, this sum extends over all motor sub-model instances assigned to that bus. Because Motor D is an algebraic performance model with no mechanical state and no inertia constant, it does not contribute stored kinetic energy; the sum is therefore restricted to Motors A, B, and C:

$$E_{k,\mathrm{load}} = \sum_{i\in\{A,\,B,\,C\}} H_i\, S_{B,i} \tag{5}$$

The per-motor MVA base $S_{B,i}$ is fixed during CLM initialisation from the model's motor-fraction parameters: if motor type i is assigned a fraction F_{m_i} of the total bus load P_{total} and operates at loading factor LF_i, then $S_{B,i} = F_{m_i}P_{\mathrm{total}}/\mathrm{LF}_i$. This is fixed at the operating point and does not re-appear in the dynamic equations; it enters (5) as a constant weight. The omission of Motor D means that (5) is a lower bound on the true stored kinetic energy of the load; the magnitude of this underestimate is characterised in Section IV.

C. Closed-Form Expression for H_{load}

The aggregate inertia constant for the load bus is obtained by normalising (5) by a chosen MVA base. Selecting $S_{\mathrm{base}} = P_{\mathrm{total}}$, where P_{total} is the total active power consumed at the bus, makes H_{load} directly comparable to synchronous generator inertia constants, which are also expressed on the machine's own rated MVA base:

$$H_{\mathrm{load}} = \frac{E_{k,\mathrm{load}}}{P_{\mathrm{total}}} = \frac{\displaystyle\sum_{i\in\{A,B,C\}} H_i\, S_{B,i}}{P_{\mathrm{total}}} \tag{6}$$

Substituting $S_{B,i} = F_{m_i}P_{\mathrm{total}}/\mathrm{LF}_i$ cancels P_{total} and leaves the aggregate inertia constant in terms of the model's own component parameters:

$$H_{\mathrm{load}} = \sum_{i\in\{A,B,C\}} \frac{F_{m_i}}{\mathrm{LF}_i}\, H_i. \tag{7}$$

Every quantity on the right-hand side — the motor fractions F_{m_i}, the loading factors LF_i, and the inertia constants H_i — is a standard composite-load-model parameter produced directly by model parameterisation. Equation (7) is therefore computable from any parameterised composite load model without additional measurement or simulation, and is linear in both the motor fractions and the inertia constants.

The non-motor fractions of the load — static (ZIP) and electronic — carry no inertia constant and so appear nowhere in the numerator of (6); they enter only through P_{total} in the denominator. Choosing P_{total} as the base therefore does *not* credit them with inertia. It normalises the motor-only kinetic energy (5) to total demand, so a bus with a large static or electronic share simply has a proportionately smaller H_{load} (since $\sum_i F_{m_i} \leq 1 - F_{\mathrm{static}} - F_{\mathrm{elec}}$). The base is a normalisation convention, not a physical energy source: the invariant physical quantity is the stored energy $E_{k,\mathrm{load}}$, and the product $H_{\mathrm{load}}\,P_{\mathrm{total}}$ returns it exactly — the P_{total} cancels the $1/P_{\mathrm{total}}$ in the definition of H_{load} — regardless of the base chosen. The choice $S_{\mathrm{base}} = P_{\mathrm{total}}$ is the one that makes H_{load} an intensive "inertia per unit of total demand," directly comparable and additive with the generator term in (8).

D. Interpretation

H_{load} represents the kinetic energy stored in the induction motor fraction of the load, normalised to the total bus load. It is a *stored-energy* (energy-content) measure: it quantifies the rotational kinetic energy physically present in the load, and thereby its capacity to transiently absorb or release energy in response to a frequency deviation — the same role played by synchronous generator inertia, and characterised by the same quantity. As made explicit below, it is an upper bound on the inertial response actually *delivered* to the grid during a disturbance.

One point of interpretation should be made explicit. The energy $E_{k,\mathrm{load}} = H_{\mathrm{load}}\,P_{\mathrm{total}}$ is referenced to synchronous speed: consistent with the definition of H in (3), it is the kinetic energy each motor would store *at* synchronous speed, which is also the convention used for synchronous generators. Because an induction motor runs at a slip s below synchronous speed, the energy physically stored at the operating point is slightly smaller, $E_{k,\mathrm{actual},i} = H_i\,S_{B,i}(1-s_i)^2$. For typical rated slips of $1\,\%$ to $5\,\%$ this is a difference of only a few percent, and adopting the synchronous-referenced value keeps the load and generation terms in (8) on a common basis; the $(1-s)^2$ correction can be applied where the operating-point energy is of direct interest.

This allows load inertia to be incorporated directly into the system-level inertia estimate. The aggregate inertia constant for a power system with synchronous generation and induction motor load is:

$$H_{\text{sys}} = \frac{H_{\text{gen}}\, S_{\text{gen}} + H_{\text{load}}\, P_{\text{total}}}{S_{\text{sys}}} \tag{8}$$

where S_{gen} is the total rated MVA of online synchronous generation and S_{sys} is the system MVA base. The load term $H_{\text{load}} P_{\text{total}}$ is the aggregate kinetic energy stored in motor loads across the system, and is directly computable from (7) without simulation. In systems with high IBR penetration, where S_{gen} is declining, this term becomes a non-negligible fraction of H_{sys} — and omitting it leads to systematic under-estimation of available inertia.

One further caveat applies to (8): it places $H_{\text{load}} P_{\text{total}}$ and $H_{\text{gen}} S_{\text{gen}}$ on a common energy basis, but a synchronous generator (rigidly coupled to system frequency) delivers that energy faster and more completely than an induction motor (coupled only through the slip–torque characteristic). Equation (7) is therefore an upper bound on the load's effective inertial contribution — distinct from, and larger than, the $(1-s)^2$ correction, which adjusts the stored energy rather than the delivered response. This coupling effect is taken up in Section VI.

E. Sensitivity and Load Electronification

Being linear in the model parameters, (7) needs no separate sensitivity analysis: H_i and LF_i are fixed weights, and the motor fractions F_{m_i} are what vary across buses and operating conditions. Two points follow. Since $\sum_i F_{m_i} \le 1$, H_{load} is bounded by $\max_i H_i/\mathrm{LF}_i$ — at most about 1 s for the NERC defaults, an order of magnitude below synchronous-generator values (3 s to 6 s). And because H_{load} scales with the aggregate motor fraction, load electronification — inverter-interfaced drives, PV, and DER displacing directly-connected motors — drives it down, the demand-side counterpart of declining synchronous inertia.

IV. Motor D and the Residential Inertia Gap

Motor D represents single-phase residential air-conditioning and refrigeration compressors [?]. Unlike Motors A–C it is a performance-based *algebraic* model — three voltage-dependent regions (run I, run II, stall) with no swing equation and hence no inertia constant, developed to reproduce fault-induced delayed voltage recovery rather than rotor dynamics [?]. It therefore contributes nothing to (5), and H_{load} omits whatever inertia the single-phase fleet carries. Had Motor D a representative inertia H_D on the same basis, its contribution would be

$$\Delta H_{\text{load}} = \frac{F_{m_D}}{\mathrm{LF}_D}\, H_D. \tag{9}$$

This bound is small: single-phase compressors are very-low-inertia machines — it is their minimal inertia that makes them stall [?] — so H_D sits below even Motor A, and the term scales with the small fraction F_{m_D} (evaluated in Section V).

More fundamentally, F_{m_D} is shrinking. Motor D was built for North American practice, where the compressor is a directly-connected single-phase capacitor-start induction motor [?]; for these the omission is a genuine gap. Modern air-conditioners instead use inverter-based variable-frequency drives [?], which the CLM rules of association assign to the *electronic* component, and whose power-electronic front end decouples the rotor from grid frequency — so zero inertia is physically correct, not a modelling artefact. The residential contribution to load inertia is thus governed by electronification (Section III-E), not by the treatment of Motor D; closing the modelling gap would recover only the small, declining directly-connected part of (9) and is outside our scope.

V. Case Study: Proposed Application to the NEM

We outline a demonstration on the Australian National Electricity Market (NEM). Rather than apply a single aggregate composition to all demand, we propose an *end-use–resolved* evaluation of (7), in which total demand is attributed across a set of end-use categories j — to be defined — each with its own rules-of-association mapping f_{ij} and an active-power share $P_j(t)$ that may vary with time:

$$\begin{aligned} H_{\text{load}}(t) &= \frac{1}{P_{\text{total}}(t)} \sum_j \sum_{i\in\{A,B,C\}} \frac{f_{ij}}{\mathrm{LF}_i}\, H_i\, P_j(t) \\ &= \sum_j p_j(t)\, h_j, \end{aligned} \tag{10}$$

where $p_j(t) = P_j(t)/P_{\text{total}}(t)$ is the power share of end use j and $h_j = \sum_{i\in\{A,B,C\}} (f_{ij}/\mathrm{LF}_i)\, H_i$ its per-end-use inertia constant. Resolving demand by end use lets categories with high directly-connected motor content — which an aggregate composition under-represents — be attributed explicitly, and lets the composition, and hence H_{load}, track diurnal and seasonal change through $P_j(t)$.

What enters the system-level estimate (8) is not the normalised constant but the load's stored kinetic energy. From (5) and the cancellation of $P_{\text{total}}(t)$ noted in Section III, this is

$$E_{\text{load}}(t) = H_{\text{load}}(t)\, P_{\text{total}}(t) = \sum_j h_j\, P_j(t), \tag{11}$$

a sum over end uses weighted by their *absolute* active power $P_j(t)$, in which $P_{\text{total}}(t)$ has cancelled against the $1/P_{\text{total}}(t)$ defining H_{load} in (10). Although P_{total} includes static- and electronic-load components that store no kinetic energy, (11) assigns them none: each h_j is built only from the motor fractions f_{ij}, so the non-inertial demand enters the normalisation base of H_{load} but never the energy E_{load}. The total demand $P_{\text{total}}(t)$ thus appears as the base on which H_{load} is reported, while $E_{\text{load}}(t)$ — the quantity carried into (8) — depends only on the directly-connected motor load.

The resulting $E_{\text{load}}(t)$ (equivalently $H_{\text{load}}(t)$) is then combined with synchronous inertia from operational dispatch to evaluate the load's contribution to system inertia via (8), over the NEM mainland synchronous area. Defining the end-use taxonomy, the mapping f_{ij} for each category, and the source of the time-varying shares $P_j(t)$ are the subject of ongoing work.

VI. Discussion

A. From Stored Energy to Delivered Response

A limitation to state plainly is that H_{load} is a stored-energy metric, not a measured frequency-response. As discussed in Section III, induction motors are coupled to system frequency far more weakly than synchronous generators — through the slip–torque characteristic rather than rigid synchronism — so the inertial energy a motor load actually delivers during a disturbance is smaller and slower than its energy content. Equation (7) therefore gives an upper bound suitable for inertia *accounting* (how much energy is present, and how it trends with electronification), but mapping it to a RoCoF contribution requires evaluating the slip dynamics of (1)–(2) under a representative disturbance. Quantifying this coupling-moderated effective inertia — and calibrating a single "response factor" relating it to H_{load} — is a natural extension and a target for measurement-based validation.

VII. Conclusion